

CARTWRIGHT, NL, Canada

WMO: 718180

Lat: **53.683N**Lon: **57.041W**Elev: **14**StdP: **101.15**Time Zone: **-4.00 (NAA)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-26.4	-24.2	-30.4	0.2	-25.8	-28.2	0.3	-23.9	19.4	-5.4	17.8	-6.5	5.3	210	0.843

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.1	24.8	17.4	22.5	16.2	20.5	15.3	18.3	23.0	17.1	21.4	15.9	19.6	4.6	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.3	11.6	20.4	15.1	10.8	19.4	13.9	9.9	18.2	51.8	23.0	48.2	21.4	44.7	19.6	26.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.9	13.2	11.2	-29.1	29.7	1.6	1.4	-30.3	30.8	-31.2	31.6	-32.2	32.4	-33.4	33.4		
(5)			-29.3	21.9	1.6	1.9	-30.4	23.2	-31.4	24.3	-32.3	25.4	-33.5	26.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	0.8	-12.8	-12.9	-8.2	-1.6	3.5	9.1	13.0	13.3	9.4	4.3	-1.3	-7.6
	DBStd	10.33	6.88	6.50	5.98	4.49	3.58	4.59	4.07	3.60	3.60	3.21	3.98	5.90
	HDD10.0	3680	706	642	564	347	206	72	17	9	55	179	339	544
	HDD18.3	6422	964	876	822	597	459	277	170	161	269	435	589	803
	CDD10.0	312	0	0	0	0	4	46	111	112	36	3	0	0
	CDD18.3	12	0	0	0	0	0	2	5	5	1	0	0	0
	CDH23.3	134	0	0	0	0	2	26	58	44	4	0	0	0
	CDH26.7	22	0	0	0	0	1	4	9	7	1	0	0	0
Wind	WSAvg	5.3	6.1	6.0	6.0	5.5	4.6	4.5	3.8	3.7	5.0	5.4	6.2	6.5
Precipitation	PrecAvg	1060	92	77	106	85	68	93	92	86	89	84	97	95
	PrecMax	1327	242	257	289	217	146	188	174	130	220	157	193	182
	PrecMin	793	23	15	27	6	22	32	37	33	43	24	43	28
	PrecStd	166	50	53	64	51	29	36	34	27	39	37	40	38
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.6	4.5	6.2	12.1	19.9	26.7	27.9	27.6	22.9	16.2	11.6	5.7
		MCWB	2.8	3.2	3.4	7.4	12.4	17.9	18.5	19.3	16.7	12.0	10.1	4.9
	2%	DB	1.2	1.5	3.6	8.8	15.6	22.8	24.9	24.1	20.0	12.9	7.7	3.1
		MCWB	0.7	1.1	1.7	5.1	9.9	15.5	17.5	17.4	15.2	10.2	6.3	2.5
	5%	DB	0.0	0.1	1.5	6.4	12.3	20.2	22.6	21.8	17.7	10.9	5.3	1.6
		MCWB	-0.3	-0.2	0.6	3.7	8.0	14.1	16.3	16.3	13.8	8.5	4.0	1.0
	10%	DB	-1.7	-2.2	0.3	4.4	9.5	17.2	20.3	19.6	15.4	9.1	3.7	0.4
		MCWB	-2.2	-2.7	-0.3	2.4	6.2	12.7	15.4	15.3	12.0	7.1	2.7	0.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.1	3.4	4.1	7.9	13.4	18.6	20.0	20.0	17.9	12.7	10.2	4.8
		MCDB	3.6	4.1	5.6	11.3	18.4	24.9	25.5	25.5	21.7	14.7	11.4	5.5
	2%	WB	0.8	1.1	2.0	5.5	10.3	16.4	18.3	18.2	16.0	10.5	6.5	2.6
		MCDB	1.2	1.4	3.1	8.2	14.9	21.4	23.0	22.6	19.2	12.6	7.6	3.0
	5%	WB	-0.3	-0.2	0.7	3.9	8.1	14.6	17.0	17.0	14.0	8.8	4.3	1.1
		MCDB	0.0	0.1	1.4	6.2	12.1	19.2	21.5	20.8	17.0	10.5	5.0	1.4
	10%	WB	-2.1	-2.6	-0.2	2.5	6.2	12.9	15.8	15.7	12.3	7.2	2.8	0.1
		MCDB	-1.6	-2.1	0.5	4.2	9.2	17.1	19.6	19.2	15.0	8.9	3.6	0.5
Mean Daily Temperature Range	5% DB	MDBR	8.0	8.7	8.6	7.3	7.5	9.3	9.8	9.1	7.6	5.3	4.9	6.0
		MCDBR	8.4	8.7	9.3	10.6	14.3	15.3	14.3	13.4	11.6	8.8	6.7	5.9
		MCWBR	8.3	8.5	8.2	7.6	9.2	9.1	8.3	7.8	7.7	6.7	6.2	5.8
	5% WB	MCDBR	8.0	8.5	8.5	9.4	13.4	14.7	12.8	12.4	11.1	8.5	6.5	5.6
		MCWBR	8.0	8.5	7.6	7.0	9.0	9.5	8.4	8.0	8.1	6.8	6.2	5.6
Clear-Sky Solar Irradiance	taub	0.255	0.277	0.296	0.338	0.370	0.376	0.405	0.370	0.342	0.315	0.279	0.261	
	taud	2.273	2.249	2.218	2.168	2.313	2.393	2.357	2.455	2.505	2.540	2.458	2.263	
	Ebn at Noon	727	826	886	887	877	874	838	852	836	787	707	635	
	Edh at Noon	58	83	107	130	119	112	114	98	82	64	50	49	
All-Sky Solar Radiation	RadAvg	0.80	1.61	2.77	4.23	4.74	4.89	4.73	4.00	2.94	1.59	0.82	0.55	
	RadStd	0.13	0.24	0.52	0.43	0.58	0.49	0.33	0.30	0.19	0.13	0.07	0.07	